

Iterated Function Systems and Hutchinson's theorem

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Introduction

- ❖ The Cantor set is made of *two* contraction mappings.
- ❖ These contractions contract by a factor of *one-third*.

C_0 	C_1 	C_2
C_3 	C_4 	C_5

Introduction

- ❖ We will define what a contraction mapping is and the definition of an IFS.

DEFINITION 2.4. Let X be a subset of $\mathbb{R}, \mathbb{R}^n$, or $\mathbb{C}$. The transformation $T : X \rightarrow X$ is called a *contraction* if there is some constant $c \in [0, 1)$ such that

$$|T(x) - T(y)| \leq c|x - y| \text{ for all } x, y \in X.$$

The number c , which is not unique, is called a *contractivity factor* for T .

DEFINITION 2.8. An *iterated function system (IFS)* is given by $\{X; T_1, T_2, \dots, T_k\}$, where each T_j is a contraction from X to itself with contractivity factor c_j . The *contractivity factor* of the IFS is given by $c = \max\{c_1, c_2, \dots, c_k\}$.

Overview

Metric spaces

Compactness

Hausdorff metric

Hutchinson's Theorem

A brief introduction to Metric Spaces

Metric spaces

The theory of metric spaces is crucial to fractal geometry

So first, we must rigorously define a metric

DEFINITION 3.1. A metric on X is a function $d : X \times X \rightarrow \mathbb{R}$ satisfying the following conditions:

- (1) $d(x, y) \geq 0$ for all $x, y \in X$,
- (2) $d(x, y) = 0$ if and only if $x = y$,
- (3) $d(x, y) = d(y, x)$ for all $x, y \in X$, and
- (4) (triangle inequality) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

Compactness for fractals

Closed sets

- ❖ Before delving deeper into metric spaces, we must define what it means for a set to be closed and bounded.
- ❖ First, we will define closed, which intuitively means a set contains all of its boundary points.

DEFINITION 3.3. We say that a set $A \subseteq X$ is *closed* if it contains its limit points. That is, if a sequence $\{a_n\} = \{a_1, a_2, a_3, \dots\}$ in A has the property that $\lim_{n \rightarrow \infty} a_n = a$, then $a \in A$. The set $A \subseteq X$ is called *open* if its complement X/A is closed.

Bounded sets

- ❖ We can now define what it means for a set to be bounded, intuitively that no portion of the set heads off to infinity.

DEFINITION 3.4. We say that a set $A \subseteq X$ is *bounded* if there is some $x_0 \in X$ and some number $r > 0$ such that $d(a, x_0) \leq r$ for all $a \in A$.

Compactness for fractals

- ❖ We can now define compactness in two ways

DEFINITION 3.5. We say that a set $A \subseteq X$ is *compact* if it is closed and bounded.

Compact!
(guaranteed by Heine-Borel Theorem) → $[0, 1]$

Cantor Intersection Theorem

- ❖ An important result that is key for proving Hutchinson's theorem that relies on compactness is the Cantor Intersection Theorem.

THEOREM 3.26. *Let $B_1, B_2, \dots$ be a sequence of nonempty compact subsets of a complete metric space such that $B_{i+1} \subseteq B_i$ for all $i = 1, 2, \dots$.*

(1) (Cantor Intersection Theorem) $\bigcap_{i=1}^{\infty} B_i$ is nonempty.

(2) $\bigcap_{i=1}^{\infty} B_i = \lim_{i \rightarrow \infty} B_i$.

Hausdorff Metric and The Space of Fractals

Space of fractals

DEFINITION 3.8. Let X be a compact subset of $\mathbb{R}^n$ or $\mathbb{C}$ with the standard Euclidean metric d . The *space of fractals* is the set

$$\mathcal{H}(X) = \{A \subseteq X \text{ such that } A \text{ is nonempty and compact}\}$$

Metric spaces - Hausdorff Metric

DEFINITION 3.9. Let A be a compact subset of X and let $\epsilon \geq 0$. The ϵ -thickening of A is the set

$$A_\epsilon = \{x \in X \text{ such that } d(x, a) \leq \epsilon \text{ for some } a \in A\}.$$

Existence of a minimum for the space of fractals

An important lemma for Hutchinson's theorem is the existence of a minimum element

LEMMA 3.11. *Let $A, B \in \mathcal{H}(X)$. Then $\min\{\epsilon \text{ such that } A \subseteq B_\epsilon\}$ exists.*

Metric spaces - Hausdorff Metric

DEFINITION 3.12. Let $A, B \in \mathcal{H}(X)$. The *Hausdorff distance* between A and B is given by

$$(3.1) \quad d_H(A, B) = \min\{\epsilon \text{ such that } A \subseteq B_\epsilon \text{ and } B \subseteq A_\epsilon\}$$

Alternatively, it is the maximum of two minimums:

$$(3.2) \quad d_H(A, B) = \max\{\min\{\epsilon \text{ s.t. } A \subseteq B_\epsilon\}, \min\{\epsilon \text{ s.t. } B \subseteq A_\epsilon\}\}$$

Hutchinson's theorem

Hutchinson's theorem

- ❖ **Key Idea:** Guarantees the existence of a Fractal through contraction mappings from iterated function systems.

THEOREM 3.25. *Let X be a compact subset of $\mathbb{R}^n$ or $\mathbb{C}$ and let $\mathcal{H}(X)$ be its space of fractals endowed with the Hausdorff metric. Let $\mathcal{T} : \mathcal{H}(X) \rightarrow \mathcal{H}(X)$ be the collage of contractions $T_1, T_2, \dots, T_k$ on X . Then*

- (1) *There exists a unique $A \in \mathcal{H}(X)$ for which $\mathcal{T}(A) = A$, and*
- (2) *For any $B \in \mathcal{H}(X)$,*

(3.5)
$$\lim_{n \rightarrow \infty} \mathcal{T}^n(B) = A$$

A quick proof to Hutchinson's Theorem

A roadmap to proving Hutchinson's Theorem (Existence)

- ❖ Consider a sequence of elements in the space of fractals.

$$T(X) \supseteq T^2(X) \supseteq \cdots \supseteq T^n(X)$$

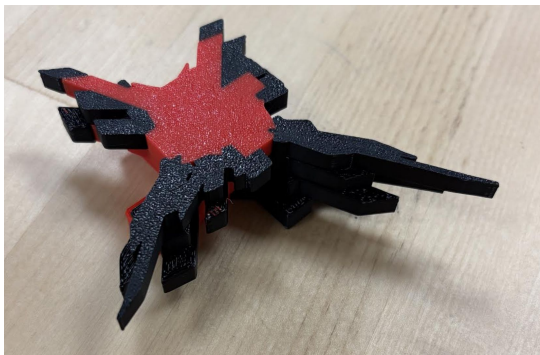
- ❖ The Cantor Intersection theorem guarantees convergence to some element **A**

$$\mathbf{A} = \bigcap_{n=1}^{\infty} T^n(X)$$

- ❖ **A** turns out to be a fixed point and is our fractal

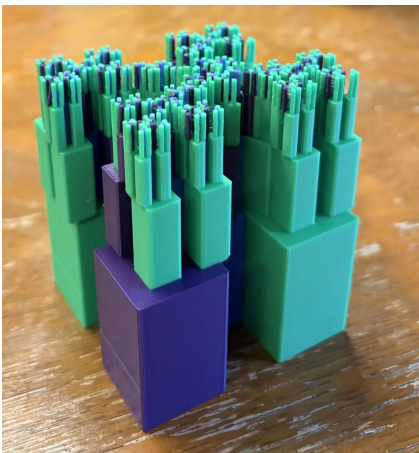
$$T(\mathbf{A}) = T\left(\lim_{n \rightarrow \infty} T^n(X)\right) = \lim_{n \rightarrow \infty} T(T^n(X)) = \lim_{n \rightarrow \infty} T^{n+1}(X) = \mathbf{A},$$

Cool Fractals that exist due to Hutchinson's theorem



Jay's Fractal Stack

**Henry's Fractal
Stack**



**Gasper's Fractal
Stack**

Thank you for your time!

Questions?

References

1. Natalie Frank, Fabricating Fractals; *Lecture Companion Notes*, Nov 2024

